Multi-view Pyramid Transformer: Look Coarser to See Broader

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Abstract

Feed-forward 3D reconstruction from many posed images demands that a model simultaneously attend to fine spatial detail within each view and maintain geometric consistency across all views. Standard transformers face a fundamental tension: full global attention suffers quadratic cost and attention dilution as the number of views grows, while purely local attention sacrifices long-range consistency. We present the ***Multi-view Pyramid Transformer (MVP)****, a scalable architecture that resolves this tension through a* dual hierarchy*: inter-view context expands progressively from individual frames, to small groups, to all views, while intra-view spatial resolution is simultaneously compressed from fine patches to coarser tokens. A lightweight convolutional token reducer between stages halves spatial resolution and doubles channel width, enlarging the effective receptive field and enabling broader view windows without quadratic blowup. A Pyramidal Feature Aggregation module inspired by DPT then fuses multi-scale features from all stages, and a lightweight decoder predicts per-pixel 3D Gaussian primitives for feed-forward rendering. On DL3DV, MVP achieves 25.96 dB PSNR with 32 input views in just 0.17 s—outperforming Long-LRM (23.97 dB, 0.84 s) and iLRM (24.30 dB, 0.53 s)—and scales seamlessly to 256 views (29.67 dB, 1.84 s) where competitors either run out of memory or degrade severely. Strong zero-shot results on Tanks&Temples and Mip-NeRF360 confirm broad generalization.*

1. Introduction

Recovering a 3D scene representation from a set of posed images is a foundational problem in computer vision. The advent of 3D Gaussian Splatting (3DGS) [19] and neural feed-forward reconstruction models [14, 17, 43, 45] has substantially lowered the cost of this operation, but most existing feed-forward methods still struggle to handle *many* high-resolution views efficiently and accurately.

The core difficulty is a fundamental tension between *local spatial fidelity* and *global geometric consistency*. Within a single image, high-resolution tokens preserve fine-grained texture and geometry. Across views, long-range attention enables cross-view correspondence and consistent 3D reasoning. Scaling a vanilla Vision Transformer [11] to V high-resolution images yields a token sequence of length $O(V \cdot HW/p^2)$, and the quadratic attention cost quickly becomes prohibitive. Beyond computation, very long token sequences suffer from *attention dilution*: individual tokens attend to an ever-larger pool of potentially irrelevant tokens, degrading reconstruction quality even when memory is available [17, 24].

Prior approaches have attacked this problem from several angles. Long-LRM [45] applies global attention at progressively downsampled resolutions but degrades sharply beyond 64 views. iLRM [17] uses an iterative refinement strategy that improves quality at the cost of latency and catastrophic degradation at high view counts. Local-attention methods [15] sidestep the quadratic cost but lose global context, leading to inconsistent reconstructions across disparate viewpoints. Linear-attention [18] and state-space [12] alternatives reduce complexity but sacrifice expressivity.

We propose a different principle: *look coarser to see broader*. As context expands from a single frame to the full scene, we simultaneously compress the spatial resolution of tokens, so that the effective computation remains tractable while global reasoning is preserved. We formalize this intuition in the **Multi-view Pyramid Transformer (MVP)**, a three-stage architecture with a dual hierarchy:

- **Inter-view hierarchy (local $\rightarrow$ group $\rightarrow$ global):** Stage 1 applies self-attention within each frame independently. Stage 2 groups g neighboring frames and performs attention within each group. Stage 3 attends globally across all views and their tokens.
- **Intra-view hierarchy (fine $\rightarrow$ coarse):** Between each stage, a strided convolutional block halves the spatial resolution of image tokens and doubles the channel width, reducing the token count by $4\times$ while expanding the receptive field.

This coupling means that by the time attention must be global, each token represents a coarse, semantically rich region of a view—keeping the sequence length manageable without discarding any view. A Pyramidal Feature Aggregation (PFA) module then fuses features from all three stages at their native resolutions, recovering spatial detail

lost during downsampling. The final decoder predicts per-pixel 3D Gaussian parameters [19] for differentiable rasterization [40].

We further incorporate Projective Positional Encoding (PRoPE) [21] as a camera-aware relative positional encoding at the group-wise and global stages, and use register tokens [10] to absorb global context without polluting patch tokens.

Extensive experiments demonstrate that MVP:

- Achieves state-of-the-art reconstruction quality across 16–128 views on DL3DV, outperforming all baselines in PSNR, SSIM, LPIPS, and inference time.
- Scales gracefully to 256 views—where Long-LRM runs out of GPU memory and iLRM degrades to 20.63 dB—maintaining 29.67 dB PSNR in under 2 seconds.
- Generalizes zero-shot to Tanks&Temples and Mip-NeRF360, outperforming iLRM by over 3 dB PSNR at 128 views.
- Supports variable view counts at test time without retraining, with consistent improvements as more views are added.

2. Related Work

Feed-forward 3D Reconstruction. LRM [14] pioneered the use of large transformer-based models for single-image-to-3D reconstruction, outputting a NeRF from a single view. Subsequent work extended this to Gaussian Splatting representations [32, 38, 43] and sparse multi-view inputs [6–8, 37]. PixelSplat [6] and MVSplat [8] handle two- to three-view inputs using epipolar priors, while DepthSplat [37] integrates monocular depth cues. These methods excel at sparse-view settings but do not scale to tens of input views.

Many-view Generalizable Reconstruction. Long-LRM [45] tackles wide-coverage reconstruction from 32–128 views by applying global attention across all tokens at progressively coarser resolutions. However, it suffers from increasing inference time and quality degradation beyond 64 views. iLRM [17] introduces iterative refinement with an alternating attention scheme that processes frames independently before merging, but its iterative nature leads to slow inference and OOM failures at very high view counts. CLiFT [35] improves multi-view consistency through cross-view lifting, focusing on the small-view regime. LVSM [16] and related methods [28] explore decoder-only architectures for novel view synthesis. DUSt3R [34] and VGGT [33] demonstrate that attention-based architectures can achieve strong 3D reasoning from many views, though they target geometric estimation rather than feed-forward rendering.

Efficient Attention for Vision. Swin Transformer [25] introduced local window attention for images. Video Swin Transformer [26] and TimeSformer [4] extended this to videos with temporal locality. These approaches restrict attention to local neighborhoods, sacrificing global context. Sparse attention patterns [9], Longformer [3], and BigBird [41] mix local and global attention for sequence modeling, but are not tailored to multi-view geometry. Token reduction approaches [5, 39] reduce token counts by merging or pruning, potentially losing spatial information. Our approach differs by hierarchically expanding inter-view context while simultaneously compressing intra-view tokens, preserving all views in the sequence.

Positional Encoding for Multi-view Transformers. Encoding camera geometry in attention has been explored through absolute ray-map encodings [45], cross-view epipolar encodings [8], and relative pose encodings [21]. PRoPE [21] encodes complete camera frustums (both intrinsics and extrinsics) as relative positional biases, improving generalization across varying view counts and intrinsics. We adopt PRoPE at our group-wise and global stages, where cross-view reasoning is most critical.

Feature Pyramid Networks. Feature pyramids [22, 42] are a standard component in dense prediction tasks, aggregating multi-scale features for improved localization. DPT [29] applied a similar idea to Vision Transformers for monocular depth estimation. We draw inspiration from DPT to design our Pyramidal Feature Aggregation module, which fuses features from all three MVP stages.

3. Method

3.1. Problem Setting and Input Representation

Given V posed images $\{\mathbf{I}_v \in \mathbb{R}^{H\times W\times 3}\}_{v=1}^{V}$ with known camera intrinsics and extrinsics, MVP predicts a set of 3D Gaussian primitives that can be rendered to novel viewpoints via differentiable rasterization.

Plücker Ray Maps. Following [21, 45], we augment each RGB image with Plücker ray maps encoding the camera geometry. For each pixel, we compute the ray origin $\mathbf{o} \in \mathbb{R}^3$ and direction $\mathbf{d} \in \mathbb{R}^3$, and form the cross product $\mathbf{o} \times \mathbf{d} \in \mathbb{R}^3$. Concatenating RGB with the full Plücker coordinates yields a 12-channel input $\mathbf{X}_v \in \mathbb{R}^{H\times W\times 12}$ per view.

Tokenization. Each 12-channel view is tokenized with a linear projection of non-overlapping $p \times p$ patches (patch size $p = 8$), followed by Layer Normalization [1]. This yields $N_1 = HW/p^2$ image tokens per view, each of dimension $d_1 = 256$.

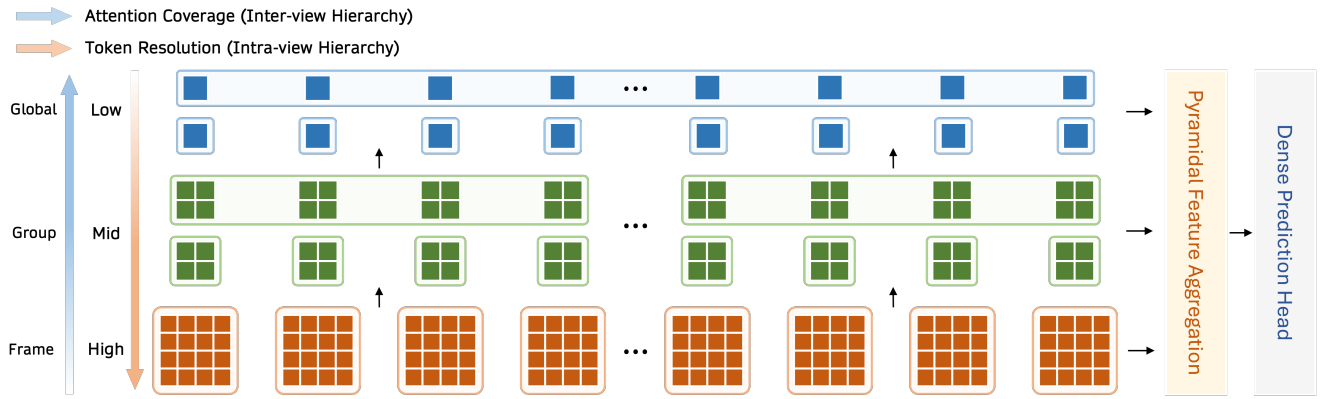

Figure 1. **Architecture Overview.** MVP processes posed input images through three stages with progressively broader inter-view context and coarser intra-view tokens. Stage 1 applies frame-wise self-attention at full patch resolution. A strided convolutional token reducer then halves the spatial resolution and doubles channel width before Stage 2 applies group-wise attention across $g = 4$ frames. A second token reducer feeds Stage 3 with global attention across all views. The Pyramidal Feature Aggregation (PFA) module fuses features from all three stages, and the Gaussian decoder predicts per-pixel 3DGS parameters for feed-forward rendering.

Register Tokens. Following [10], we prepend $R = 4$ learnable register tokens to each view's token sequence before Stage 1. Register tokens accumulate global context without contaminating local patch representations, and their aggregated information is propagated to subsequent stages via a learned projection.

3.2. Three-Stage Dual Hierarchy

MVP processes tokens through three transformer stages with a progressively expanding inter-view attention scope and a simultaneously shrinking intra-view token resolution. Figure 1 illustrates the overall pipeline.

Stage 1: Frame-wise Attention. Each view's tokens (register + image patches) are processed independently. With token sequence length $L_1 = R + N_1$, Stage 1 applies $n_1 = 2$ transformer blocks at dimension $d_1 = 256$. The quadratic cost is $O(V \cdot L_1^2)$, linear in the number of views. Standard scaled dot-product attention with QK-normalization [13] is used.

Token Reducer 1. After Stage 1, we split the register tokens from the image patch tokens. Register tokens are projected to dimension $d_2 = 512$ via a learned linear layer for propagation to Stage 2. Image tokens of shape $(HW/p^2, d_1)$ are reshaped to a spatial feature map of size $(H/p) \times (W/p)$ and passed through a strided depth-wise convolutional block with kernel size 2×2 and stride 2, reducing the spatial dimensions to $(H/2p) \times (W/2p)$ while expanding channels from d_1 to d_2. This halves the per-view token count from N_1 to $N_2 = N_1/4$.

Stage 2: Group-wise Attention. Views are arranged into non-overlapping groups of $g = 4$ consecutive views. Within each group, register tokens and image tokens from all g views are concatenated into a single sequence of length $L_2 = R + g \cdot N_2$ and processed by $n_2 = 4$ transformer blocks at dimension $d_2 = 512$. This intermediate level of inter-view attention—bridging purely frame-local and fully global attention—allows each view to integrate context from its nearest spatial neighbors before committing to global reasoning.

We apply Projective Positional Encoding (PRoPE) [21] at Stage 2 to condition attention on relative camera geometry within each group. PRoPE encodes both intrinsic and extrinsic parameters as relative attention biases, enabling the model to reason about epipolar geometry without explicit geometric computation.

Token Reducer 2. Analogously to Token Reducer 1, image tokens are reshaped and passed through a second strided convolutional block, halving the spatial resolution to $N_3 = N_2/4 = N_1/16$ tokens per view at dimension $d_3 = 1024$.

Stage 3: Global Attention. All V views are combined into a single sequence of length $L_3 = R + V \cdot N_3$, and $n_3 = 8$ transformer blocks at dimension $d_3 = 1024$ perform global self-attention across the full scene. At this stage, each token represents a $4p \times 4p$ region of the original image, enabling efficient global reasoning without quadratic blowup. PRoPE is again applied at Stage 3, now conditioning on camera geometry across the full view set.

3.3. Computational Analysis

The total attention cost of MVP scales as:

$$\mathcal{O}\left(VL_1^2 + \frac{V}{g}L_2^2 + L_3^2\right) \approx \mathcal{O}(VN_1^2 + VN_1^2/g + V^2N_1^2/256), \tag{1}$$

where the last term is the global cost, reduced by $256\times$ relative to full-resolution global attention. In contrast, global attention at full resolution would cost $\mathcal{O}(V^2N_1^2)$—$256\times$ more expensive at Stage 3 alone. This analysis ignores register tokens ($R \ll N_v$) for clarity.

3.4. Pyramidal Feature Aggregation

After Stage 3, we extract per-view image tokens from all three stages and aggregate them using a Pyramidal Feature Aggregation (PFA) module inspired by DPT [29]. Specifically, Stage 1 tokens at resolution $(H/p, W/p)$ are normalized and projected; Stage 2 tokens at resolution $(H/2p, W/2p)$ are similarly processed; Stage 3 tokens at resolution $(H/4p, W/4p)$ are upsampled back to $(H/2p, W/2p)$ and fused via addition with Stage 2 features, then further upsampled and fused with Stage 1 features. The resulting feature map retains the spatial resolution of Stage 1 while incorporating global context from Stage 3. This design recovers spatial detail that would otherwise be lost due to the progressive downsampling, and provides a rich multi-scale representation for Gaussian prediction.

3.5. 3D Gaussian Decoder

The aggregated feature map from PFA, of shape (BV, N_1, d_3), is passed through a lightweight decoder consisting of a Layer Normalization and a single linear layer, predicting per-patch Gaussian parameters. Following [43], we predict the following per-pixel attributes, which are then unpacked from patch space:

- **Position offset** $\Delta\mathbf{x} \in \mathbb{R}^3$: The Gaussian center is computed as $\mathbf{x} = \sigma(\Delta x) \cdot d_{\max} \cdot \hat{\mathbf{d}} + \mathbf{o}$, where $\hat{\mathbf{d}}$ is the pixel ray direction and $d_{\max} = 500$.
- **Color features** $\mathbf{f} \in \mathbb{R}^{3(d_{sh}+1)^2}$: Spherical harmonic (SH) coefficients with degree $d_{sh} = 1$ for view-dependent appearance.
- **Scale** $\mathbf{s} \in \mathbb{R}^3$: Log-scale anisotropic covariance.
- **Rotation** $\mathbf{q} \in \mathbb{R}^4$: Unit quaternion.
- **Opacity** $o \in \mathbb{R}^{(d_{op}+1)^2}$: View-dependent opacity via SH coefficients with degree $d_{op} = 2$, enabling accurate handling of transparent regions.

The predicted Gaussians are rendered to target views using differentiable rasterization via gsplat [40].

3.6. Training

Training Protocol. We train MVP in two stages. Stage 1 trains for 100K steps at 480×270 resolution with 32 input views, batch size 8 per GPU, using an L2 reconstruction loss and a perceptual loss [42] with weights $\lambda_{L2} = 1.0$ and $\lambda_{\text{perc}} = 0.2$. Stage 2 fine-tunes for 50K additional steps at 960×544 resolution with frozen Stage 1 and Stage 2 parameters and a lower learning rate. We use AdamW [27] with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.05, and BF16 automatic mixed precision. Gradient checkpointing is applied at every block.

Opacity Regularization. To prevent degenerate opacity solutions, we regularize with a random-direction opacity loss: we compute SH-decoded opacity at uniformly random directions and penalize its mean with weight $\lambda_{\text{op}} = 10^{-3}$.

Variable View Count. Unlike methods that train at a fixed view count, MVP is trained with a fixed number of 32 views but evaluated across 16–256 views by simply adjusting the grouping and global attention to the available views. No retraining or fine-tuning is required for different view counts, enabling strong generalization.

4. Experiments

4.1. Experimental Setup

Datasets. We train on DL3DV-10K [23], a large-scale dataset of 10K real-world scenes with SfM-recovered camera poses [30, 31]. We split 8 scenes per-fold for evaluation following iLRM [17]. We also evaluate on RE10K [44] for indoor novel view synthesis. For zero-shot evaluation, we use Tanks&Temples [20] and Mip-NeRF360 [2].

Baselines. We compare against Long-LRM [45] and iLRM [17]—the two main feed-forward multi-view reconstruction models supporting many-view input—and CLiFT [35] on RE10K. We also include per-scene optimized 3D-GS [19] (30K iterations) as an upper-bound reference.

Metrics. We report PSNR (↑), SSIM (↑), and LPIPS (↓) for reconstruction quality, and inference time (seconds) on a single NVIDIA H100 80 GB GPU.

4.2. Main Results on DL3DV

Table 1 presents quantitative comparisons on DL3DV for 16–128 input views.

MVP consistently outperforms all baselines across all view settings. At 32 views, MVP achieves 25.96 dB PSNR and 0.187 LPIPS in only 0.17 s, compared to Long-LRM's 23.97 dB in 0.84 s and iLRM's 24.30 dB in 0.53 s—representing a $5\times$ speedup over Long-LRM while improving PSNR by nearly 2 dB. At 128 views, MVP (29.02 dB, 0.77 s) approaches per-scene 3D-GS quality (29.43 dB, 8 min) at a fraction of the cost, while both Long-LRM (21.24

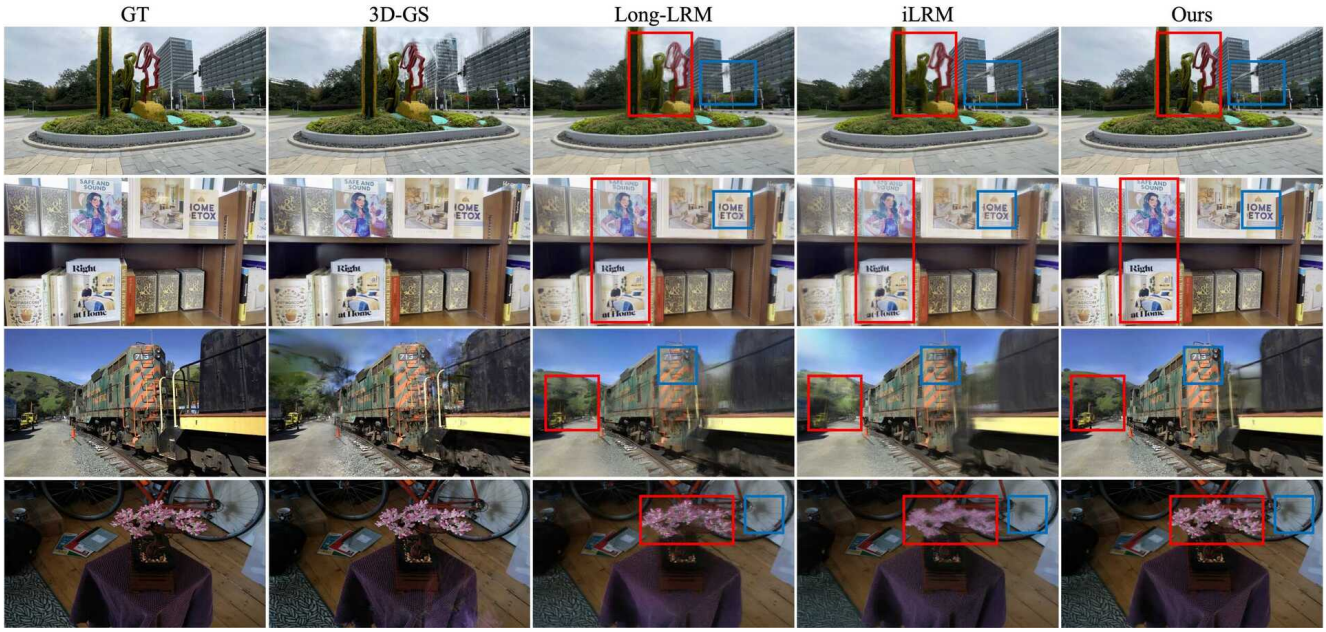

Figure 2. **Qualitative results.** Top two rows: DL3DV scenes reconstructed by MVP (32 input views). Third row: Tanks&Temples (128 views, zero-shot). Bottom row: Mip-NeRF360 (128 views, zero-shot). MVP faithfully captures fine details and handles diverse lighting conditions, indoor/outdoor scenes, and large-scale unbounded environments.

Method	16 views				32 views				64 views				128 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [19]	21.48	0.753	0.252	8min	24.43	0.827	0.191	8min	27.34	0.883	0.146	8min	29.43	0.914	0.123	8min
Long-LRM [45]	21.05 (22.66)	0.708 (0.740)	0.297 (0.292)	0.50	23.97	0.778	0.267	0.84	23.60	0.789	0.260	2.08	21.24	0.739	0.308	6.39
iLRM [17]	21.92 (22.91)	0.748 (0.766)	0.316 (0.295)	0.19	24.30	0.803	0.256	0.53	24.44	0.819	0.240	1.66	22.98	0.807	0.249	5.61
Ours	**23.76**	**0.798**	**0.239**	**0.09**	**25.96**	**0.847**	**0.187**	**0.17**	**27.73**	**0.881**	**0.154**	**0.36**	**29.02**	**0.903**	**0.134**	**0.77**

Table 1. Quantitative comparisons on the DL3DV dataset with varying numbers of input views. The values in parentheses $(\cdot)$, for the 16-view setting are taken from the original papers, where the models are trained with a fixed 16-view input. For all metrics, we report the results by re-evaluating the models from their 32-view checkpoints.

dB) and iLRM (22.98 dB) degrade significantly due to attention dilution.

Notably, MVP exhibits consistent quality improvement as more views are added—a behavior that is absent in Long-LRM and iLRM, which peak around 32–64 views and deteriorate thereafter. This demonstrates that MVP effectively leverages additional context without being overwhelmed by long sequences.

4.3. Extreme View Counts: 192 and 256 Views

Method	192 views				256 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [19]	30.14	0.924	0.116	8min	30.39	0.926	0.114	8min
Long-LRM [45]	OOM (GPU memory limit exceeded, 80 GB)							
iLRM [17]	21.68	0.788	0.264	11.91	20.63	0.767	0.281	20.92
Ours	**29.54**	**0.912**	**0.127**	**1.23**	**29.67**	**0.915**	**0.128**	**1.84**

Table 2. Quantitative comparisons on the DL3DV dataset with varying numbers of input views.

Table 2 extends results to 192 and 256 views. Long-LRM runs out of GPU memory (80 GB) at these settings, while iLRM degrades to 20.63 dB at 256 views in over 20 seconds. MVP maintains 29.67 dB PSNR at 256 views in just 1.84 s—approaching per-scene 3D-GS quality at over 260× less time. This dramatic efficiency advantage stems directly from the dual-hierarchy design: by the time all 256 views must communicate globally, each view is represented by only $N_3 = N_1/16$ tokens at high channel width, keeping global attention tractable.

4.4. Zero-shot Generalization

Table 3 shows zero-shot results on Tanks&Temples [20] and Mip-NeRF360 [2]. MVP consistently outperforms baselines across all view counts (32, 64, 128). At 128 views, MVP achieves 22.36 dB on Tanks&Temples and 25.12 dB on Mip-NeRF360, surpassing iLRM by 3.14 dB and 3.80 dB respectively. Crucially, MVP continues to improve with more views on both datasets, while iLRM degrades at 128

Method	Views	Tanks & Temples			Mip-NeRF360		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Long-LRM [45]		18.59	0.614	0.366	21.08	0.484	0.445
iLRM [17]	32	18.58	0.631	0.385	21.09	0.495	0.466
Ours		**19.54**	**0.708**	**0.277**	**22.21**	**0.587**	**0.355**
Long-LRM		19.44	0.651	0.334	21.30	0.499	0.431
iLRM	64	19.82	0.692	0.318	21.60	0.522	0.444
Ours		**21.24**	**0.761**	**0.221**	**23.72**	**0.656**	**0.302**
Long-LRM		18.47	0.613	0.375	19.82	0.484	0.457
iLRM	128	19.22	0.696	0.319	21.32	0.551	0.424
Ours		**22.36**	**0.804**	**0.184**	**25.12**	**0.736**	**0.248**

Table 3. Quantitative comparisons on the Tanks&Temples and Mip-NeRF360 under a zero-shot setting. For the 128 view setting on Mip-NeRF360, the `stump` and `treehill` scenes are excluded due to their limited number of frames.

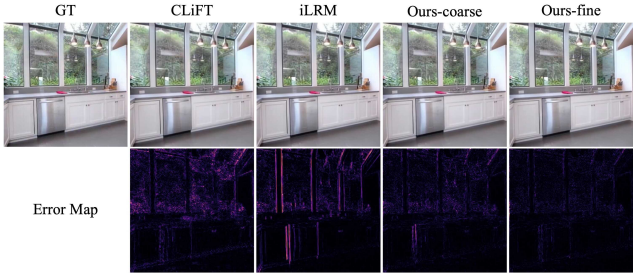

Figure 3. **Qualitative comparison on RE10K (4 views).** Error maps (right) show that MVP-fine recovers sharper edges and finer texture than CLiFT and iLRM.

views compared to 64. This confirms that MVP's hierarchical design prevents the attention dilution that plagues prior approaches.

4.5. RE10K Results

Method	4 views			8 views		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CLiFT [35]	30.13	0.916	0.095	29.68	0.910	0.092
iLRM [17]	30.37	0.923	0.095	28.90 (31.57)	0.920 (0.935)	0.102 (0.082)
Ours-coarse	30.56	0.925	0.087	31.78	0.939	0.077
Ours-fine	**32.12**	**0.941**	**0.077**	**33.40**	**0.952**	**0.066**

Table 4. Quantitative comparisons on the RE10K dataset. Baselines are evaluated using 4-view models, and values in parentheses $(\cdot)$ are from the models trained for that specific view setting.

Table 4 compares on RE10K. We report two variants: MVP-coarse (using Stage-3 coarse tokens only) and MVP-fine (using full PFA aggregation). MVP-fine achieves 33.40 dB PSNR with 8 views, surpassing CLiFT (29.68 dB) and iLRM (28.90 dB) by substantial margins. Even MVP-coarse (31.78 dB) outperforms all baselines. The large improvement from coarse to fine (1.62 dB at 8 views) confirms that the Pyramidal Feature Aggregation module is critical for recovering spatial detail needed in high-quality indoor scene synthesis.

4.6. Longer Context Generalization

Table 5 evaluates generalization to view counts unseen during training (40 and 48 views, trained on 32). MVP

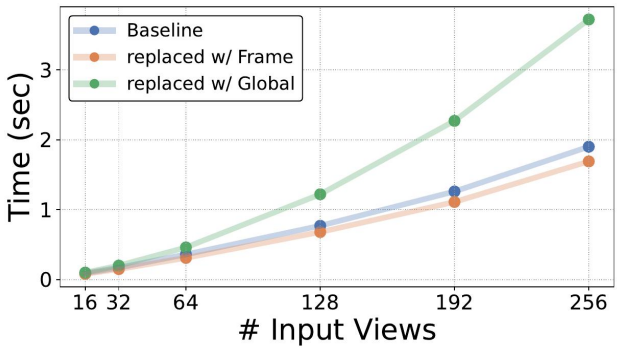

Figure 4. **Ablation analysis.** PSNR vs. inference time for key MVP design choices: removing group-wise attention and varying the attention hierarchy design. MVP achieves the best accuracy-speed trade-off.

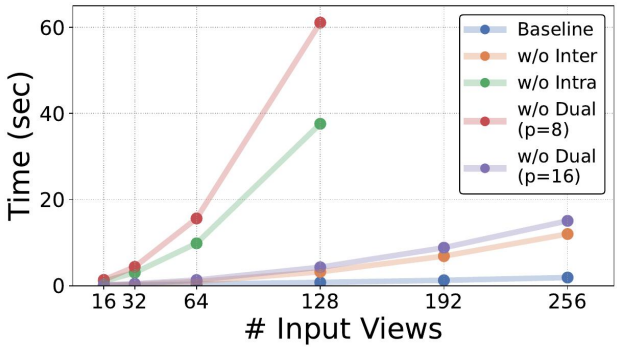

Figure 5. **Hierarchy ablation.** Comparison of dual hierarchy, inter-view-only hierarchy, intra-view-only hierarchy, and no hierarchy. The dual design yields the best PSNR at all view counts.

smoothly handles unseen context lengths, gaining 0.48 dB at 40 views and 1.18 dB at 48 views relative to the 32-view baseline. In contrast, Long-LRM and iLRM show smaller relative improvements, indicating that their attention mechanisms do not generalize as gracefully to out-of-distribution lengths. This robustness arises from the dual-hierarchy design: adding more views simply extends the group-wise and global attention sequences, without fundamentally changing the computational structure.

5. Ablation and Analysis

5.1. Component Ablations

Table 6 presents ablation results. All variants are trained at 256×256 resolution for 100K iterations on DL3DV.

Role of Pyramidal Feature Aggregation. Removing the PFA module ("w/o Feature Agg.") causes a dramatic drop from 22.79 to 21.58 dB PSNR (and LPIPS from 0.235 to 0.340), underscoring its importance for recovering spatial

Method	32 views (Trained)				40 views (Unseen)				48 views (Unseen)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Long-LRM [45]	23.97	0.778	0.267	0.84	24.18 (0.21)	0.787	0.260	1.05 (0.21)	24.30 (0.33)	0.797	0.252	1.38 (0.54)
iLRM [17]	24.30	0.803	0.257	0.53	24.54 (0.24)	0.811	0.248	0.76 (0.23)	24.78 (0.48)	0.820	0.240	1.04 (0.51)
Ours	**25.88**	**0.845**	**0.188**	**0.17**	**26.36** (0.48)	**0.855**	**0.178**	**0.22** (0.05)	**27.06** (1.18)	**0.870**	**0.164**	**0.26** (0.09)

Table 5. Longer context generalization. Values in parentheses $(\cdot)$ denote the improvements over 32 views.

Method	Stage 1	Stage 2	Stage 3	Patch Size	Hidden Dim.	Feature Agg.	PSNR ↑	SSIM ↑	LPIPS ↓
Baseline	Frame (2)	Group (4)	Global (8)	(8, 16, 32)	(128, 256, 512)	o	22.79	0.733	0.235
w/o Feature Agg.	Frame (2)	Group (4)	Global (8)	(8, 16, 32)	(128, 256, 512)	x	21.58	0.646	0.340
w/o Group-attn.	Frame (2)	Frame (4)	Global (8)	(8, 16, 32)	(128, 256, 512)	o	22.53	0.720	0.247
	Frame (2)	Global (4)	Global (8)				22.94	0.739	0.235
w/o Inter-view Hierarchy	Global (2)	Global (4)	Global (8)	(8, 16, 32)	(128, 256, 512)	o	22.94	0.739	0.236
w/o Intra-view Hierarchy	Frame (2)	Group (4)	Global (8)	(8, 8, 8)	(512, 512, 512)	o	22.83	0.732	0.249
w/o Dual Hierarchy (p=8)	Global (2)	Global (4)	Global (8)	(8, 8, 8)	(512, 512, 512)	o	23.20	0.747	0.241
w/o Dual Hierarchy (p=16)	Global (2)	Global (4)	Global (8)	(16, 16, 16)	(512, 512, 512)	o	21.80	0.651	0.341
Reversed Hierarchy	Global (8)	Group (4)	Frame (2)	(32, 16, 8)	(512, 256, 128)	o	18.95	0.442	0.555

Table 6. Ablations. We train all variants for 100K iterations at a reduced resolution (256×256) on the DL3DV dataset.

detail from coarse Stage 3 features.

Group-wise Attention. Replacing Stage 2 group-wise attention with either extended frame-wise attention ("Frame (4)") or immediate global attention ("Global (4)") both degrade performance relative to the full baseline. "Frame (4)" restricts cross-view context entirely at this stage (22.53 dB), while jumping directly to global attention (22.94 dB) skips the intermediate bridging level. The group-wise stage provides a critical intermediate scope that bridges frame-local and scene-global reasoning.

Inter-view Hierarchy. "w/o Inter-view Hierarchy"—which replaces all three stages with global attention—achieves 22.94 dB but at higher computational cost, as global attention must handle both coarse and fine tokens. The hierarchy permits fine-grained local processing at early stages where it is most beneficial.

Intra-view Hierarchy. "w/o Intra-view Hierarchy"—which keeps all stages at the same resolution (patch size 8, channel 512 throughout)—reduces to 22.83 dB, showing that the progressive token compression is also necessary. Without it, global attention at Stage 3 must handle the full resolution token sequence, causing both efficiency and attention dilution issues.

Dual Hierarchy. Removing both hierarchies simultaneously ("w/o Dual Hierarchy") with a fixed patch size of 8 surprisingly achieves 23.20 dB—slightly above the baseline with all hierarchies. However, this comes at substantially higher computational cost (all stages are 512-channel global attention). With patch size 16, performance collapses to 21.80 dB, confirming that the dual hierarchy is essential for retaining quality at larger patch sizes.

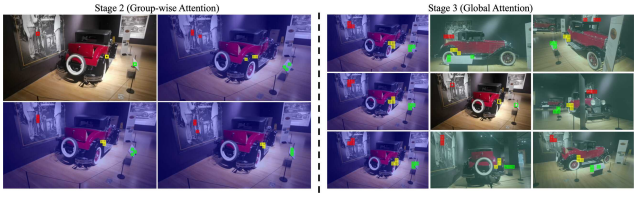

Figure 6. **Attention visualization.** Stage 1 attends locally within each frame (left). Stage 2 group-wise attention focuses on geometrically consistent regions across neighboring views (center). Stage 3 global attention captures long-range correspondences across disparate viewpoints (right). The progressive expansion of attention scope is clearly visible.

Reversed Hierarchy. A model with the reversed order (global attention first, then group, then frame, with correspondingly fine resolution at Stage 3) collapses to 18.95 dB PSNR—confirming that local-to-global ordering with simultaneous fine-to-coarse compression is critical and cannot be reversed.

5.2. Attention Visualization

Figure 6 visualizes attention maps from a query pixel across the three stages. Stage 1 exhibits sharp, local attention within the source frame, focusing on nearby pixels with similar appearance. Stage 2 group-wise attention extends across the group's views, concentrating on geometrically co-visible regions. Stage 3 global attention spans all views and captures long-range correspondences, including views on the opposite side of the scene.

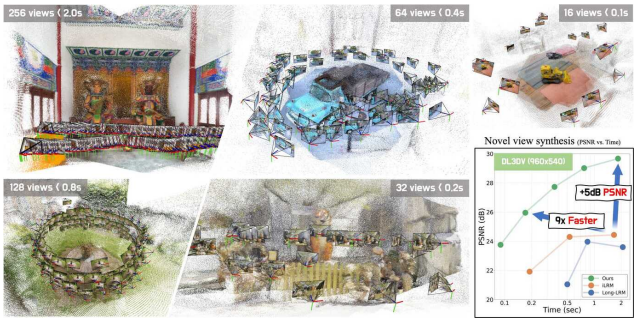

Figure 7. **MVP handles 16–256 views** in a single forward pass, reconstructing diverse large-scale scenes in under 0.1–2.0 seconds. Reconstruction quality consistently improves with more views.

This progressive pattern confirms that the dual hierarchy naturally encourages appropriate attention patterns at each level: fine-grained local processing early on, coarse global reasoning later. The coarser token resolution at Stage 3 acts as a natural regularizer, preventing attention from being wasted on irrelevant fine-grained details in distant views.

5.3. Teaser: Scalability across View Counts

Figure 7 demonstrates MVP's scalability across a wide range of view counts. Unlike prior methods that degrade or fail at high view counts, MVP reliably improves quality and handles diverse scene types (indoor, outdoor, bounded, unbounded) with increasing views.

6. Conclusion and Discussions

We presented **MVP**, a Multi-view Pyramid Transformer for scalable feed-forward 3D reconstruction from many posed images. The key insight—*look coarser to see broader*—drives a dual hierarchy that couples progressive inter-view context expansion with simultaneous intra-view token compression. By the time all views communicate globally, each token represents a coarse, information-dense region, keeping the sequence length tractable without sacrificing any view. A Pyramidal Feature Aggregation module recovers spatial detail for dense Gaussian prediction, and PRoPE provides camera-aware relative positional encoding at the inter-group and global stages.

MVP achieves state-of-the-art results across 16–256 views on DL3DV, with strong zero-shot generalization to Tanks&Temples, Mip-NeRF360, and RE10K. It reconstructs scenes from 32 views in 0.17 seconds and from 256 views in 1.84 seconds on a single H100 GPU, dramatically outpacing alternatives that either run out of memory or degrade at high view counts.

Limitations. MVP requires pre-computed camera poses, limiting applicability to pose-free settings. The current training resolution is fixed; handling variable-resolution inputs at test time may require architectural adjustments. Future work could explore joint pose estimation and reconstruction, or extend the dual hierarchy to video sequences for dynamic scene reconstruction [36].

Broader Impact. Fast, high-quality 3D reconstruction from many views has applications in AR/VR, autonomous driving, robotics, and digital heritage preservation. As with any reconstruction system, malicious use to reconstruct private spaces from internet imagery is a potential concern. We encourage responsible deployment.

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

[2] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. Mip-nerf 360: Unbounded anti-aliased neural radiance fields. *CVPR*, 2022.

[3] Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *Icml*, page 4, 2021.

[5] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your vit but faster. *arXiv preprint arXiv:2210.09461*, 2022.

[6] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19457–19467, 2024.

[7] Anpei Chen, Haofei Xu, Stefano Esposito, Siyu Tang, and Andreas Geiger. Lara: Efficient large-baseline radiance fields. In *European Conference on Computer Vision*, pages 338–355. Springer, 2024.

[8] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. Mvsplat: Efficient 3d gaussian splatting from sparse multi-view images. In *European Conference on Computer Vision*, pages 370–386. Springer, 2024.

[9] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.

[10] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. *arXiv preprint arXiv:2309.16588*, 2023.

[11] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

[12] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First conference on language modeling*, 2024.

[13] Alex Henry, Prudhvi Raj Dachapally, Shubham Pawar, and Yuxuan Chen. Query-key normalization for transformers. *arXiv preprint arXiv:2010.04245*, 2020.

[14] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3d. In *ICLR*, 2024.

[15] Tooba Imtiaz, Lucy Chai, Kathryn Heal, Xuan Luo, Jungyeon Park, Jennifer Dy, and John Flynn. Lvt: Large-scale scene reconstruction via local view transformers. *arXiv preprint arXiv:2509.25001*, 2025.

[16] Haian Jin, Hanwen Jiang, Hao Tan, Kai Zhang, Sai Bi, Tianyuan Zhang, Fujun Luan, Noah Snavely, and Zexiang Xu. Lvsm: A large view synthesis model with minimal 3d inductive bias. In *ICLR*, 2025.

[17] Gyeongjin Kang, Seungtae Nam, Seungkwon Yang, Xiangyu Sun, Sameh Khamis, Abdelrahman Mohamed, and Eunbyung Park. ilrm: An iterative large 3d reconstruction model. *arXiv preprint arXiv:2507.23277*, 2025.

[18] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning*, pages 5156–5165. PMLR, 2020.

[19] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023.

[20] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM Transactions on Graphics*, 36(4), 2017.

[21] Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. *Advances in Neural Information Processing Systems*, 2025.

[22] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2117–2125, 2017.

[23] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. Dl3dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22160–22169, 2024.

[24] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12: 157–173, 2024.

[25] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[26] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video swin transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3202–3211, 2022.

[27] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[28] Nithin Gopalakrishnan Nair, Srinivas Kaza, Xuan Luo, Vishal M Patel, Stephen Lombardi, and Jungyeon Park. Scaling transformer-based novel view synthesis with models token disentanglement and synthetic data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 28567–28576, 2025.

[29] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021.

[30] Johannes Lutz Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[31] Johannes Lutz Schönberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. Pixelwise view selection for unstructured multi-view stereo. In *European Conference on Computer Vision (ECCV)*, 2016.

[32] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. In *European Conference on Computer Vision*, pages 1–18. Springer, 2024.

[33] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025.

[34] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

[35] Zhengqing Wang, Yuefan Wu, Jiacheng Chen, Fuyang Zhang, and Yasutaka Furukawa. Clift: Compressive light-field tokens for compute-efficient and adaptive neural rendering. *arXiv preprint arXiv:2507.08776*, 2025.

[36] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4d gaussian splatting for real-time dynamic scene rendering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20310–20320, 2024.

[37] Haofei Xu, Songyou Peng, Fangjinhua Wang, Hermann Blum, Daniel Barath, Andreas Geiger, and Marc Pollefeys. Depthsplat: Connecting gaussian splatting and depth. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 16453–16463, 2025.

[38] Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and Gordon Wetzstein. Grm: Large gaussian reconstruction model for efficient 3d reconstruction and generation. In *European Conference on Computer Vision*, pages 1–20. Springer, 2024.

[39] Senqiao Yang, Yukang Chen, Zhuotao Tian, Chengyao Wang, Jingyao Li, Bei Yu, and Jiaya Jia. Visionzip: Longer

is better but not necessary in vision language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19792–19802, 2025.

[40] Tianzhu Ye, Li Dong, Yuqing Xia, Yutao Sun, Yi Zhu, Gao Huang, and Furu Wei. Differential transformer. *arXiv preprint arXiv:2410.05258*, 2024.

[41] Manzil Zaheer, Guru Guruganesh, Kumar Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, et al. Big bird: Transformers for longer sequences. *Advances in neural information processing systems*, 33:17283–17297, 2020.

[42] Dong Zhang, Hanwang Zhang, Jinhui Tang, Meng Wang, Xiansheng Hua, and Qianru Sun. Feature pyramid transformer. In *European conference on computer vision*, pages 323–339. Springer, 2020.

[43] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. Gs-lrm: Large reconstruction model for 3d gaussian splatting. In *European Conference on Computer Vision*, pages 1–19. Springer, 2024.

[44] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: Learning view synthesis using multiplane images. *arXiv preprint arXiv:1805.09817*, 2018.

[45] Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu. Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4349–4359, 2025.